

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

ESP32-C6 radio



File: radio_esp32.kicad_sch

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

Footprint and PCB qualification are deferred by scope.

RA8P1 IO and analog supplies



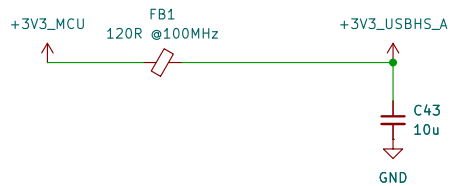
File: mcu_supply_power.kicad_sch

RA8P1 clocks, reset and debug

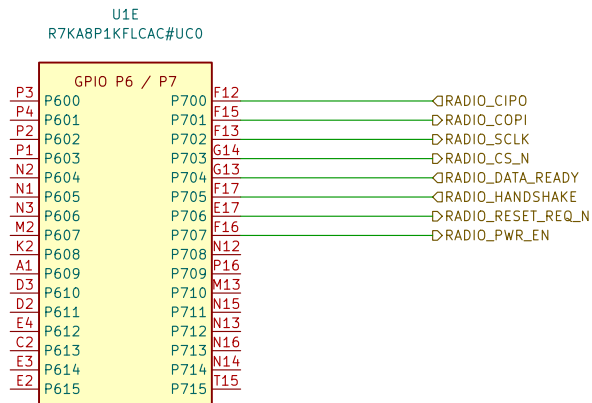
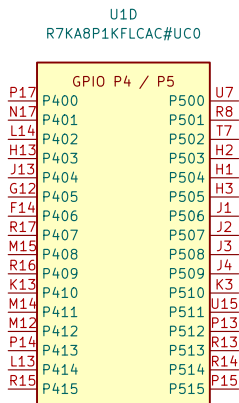
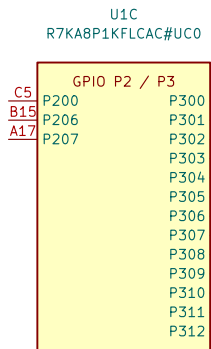
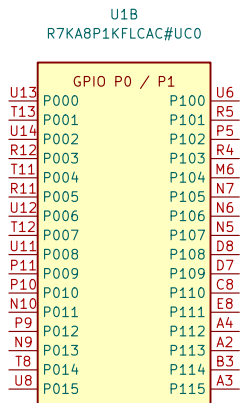


File: mcu_clocks_debug.kicad_sch

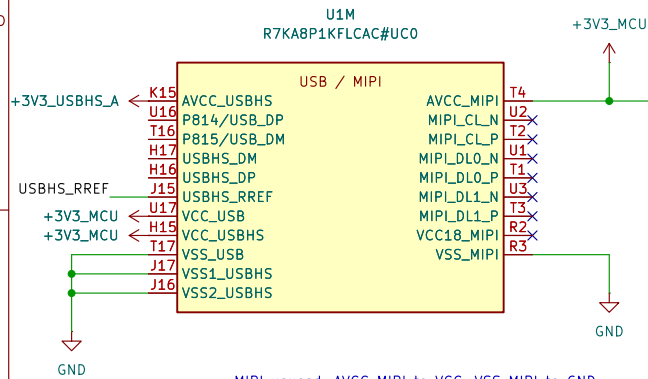
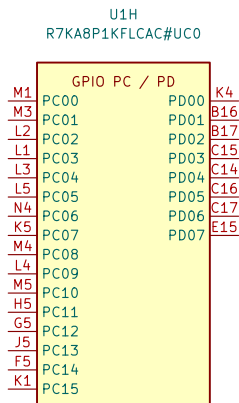
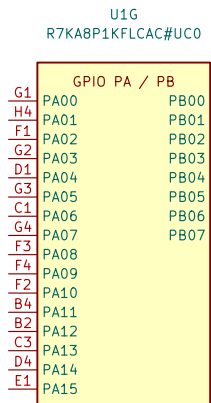
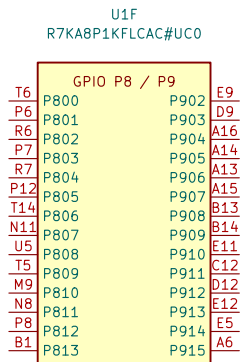
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



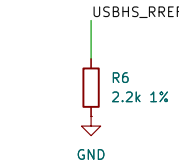
USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm; 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]



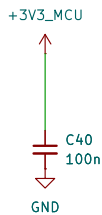
RADIO-008: SPIA_C P700..P703; IRQ26=P704, IRQ19=P705.
P706=RESET_REQ_N; P707=PWR_EN. Host-domain signals.
Switched-domain isolation and reset arbitration remain open.
Pin map + conflicts: ../design/radio_interface.md#radio-008-ra8p1-host-allocation



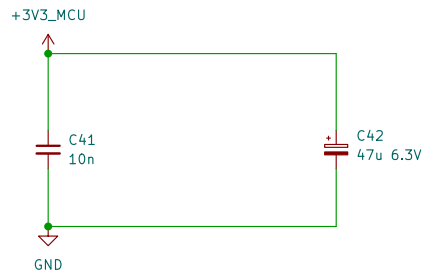
MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.



USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md



USB-002: USB full-speed supply / U1,U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md



USB-003: USB-HS supply / U1,H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

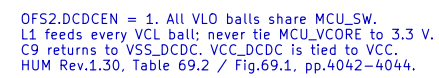
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Date: 2026-09-05

Rev: 0.3 DRAFT

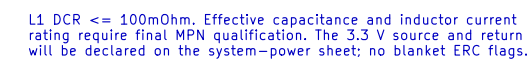
KiCad E.D.A. 10.0.5

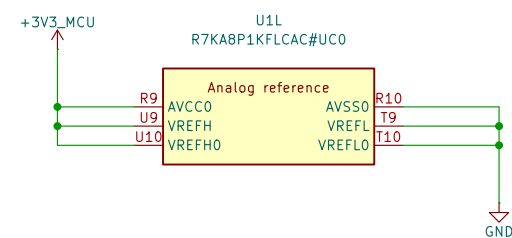
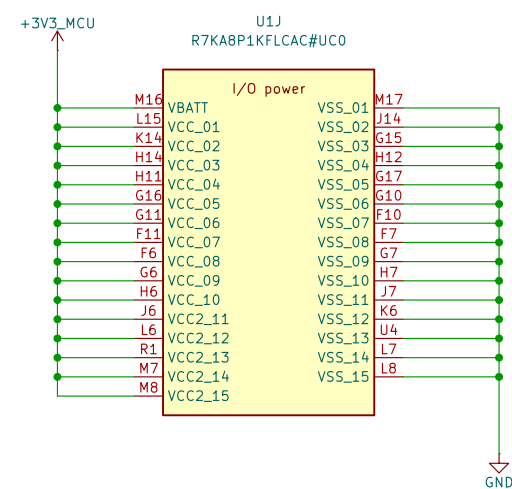
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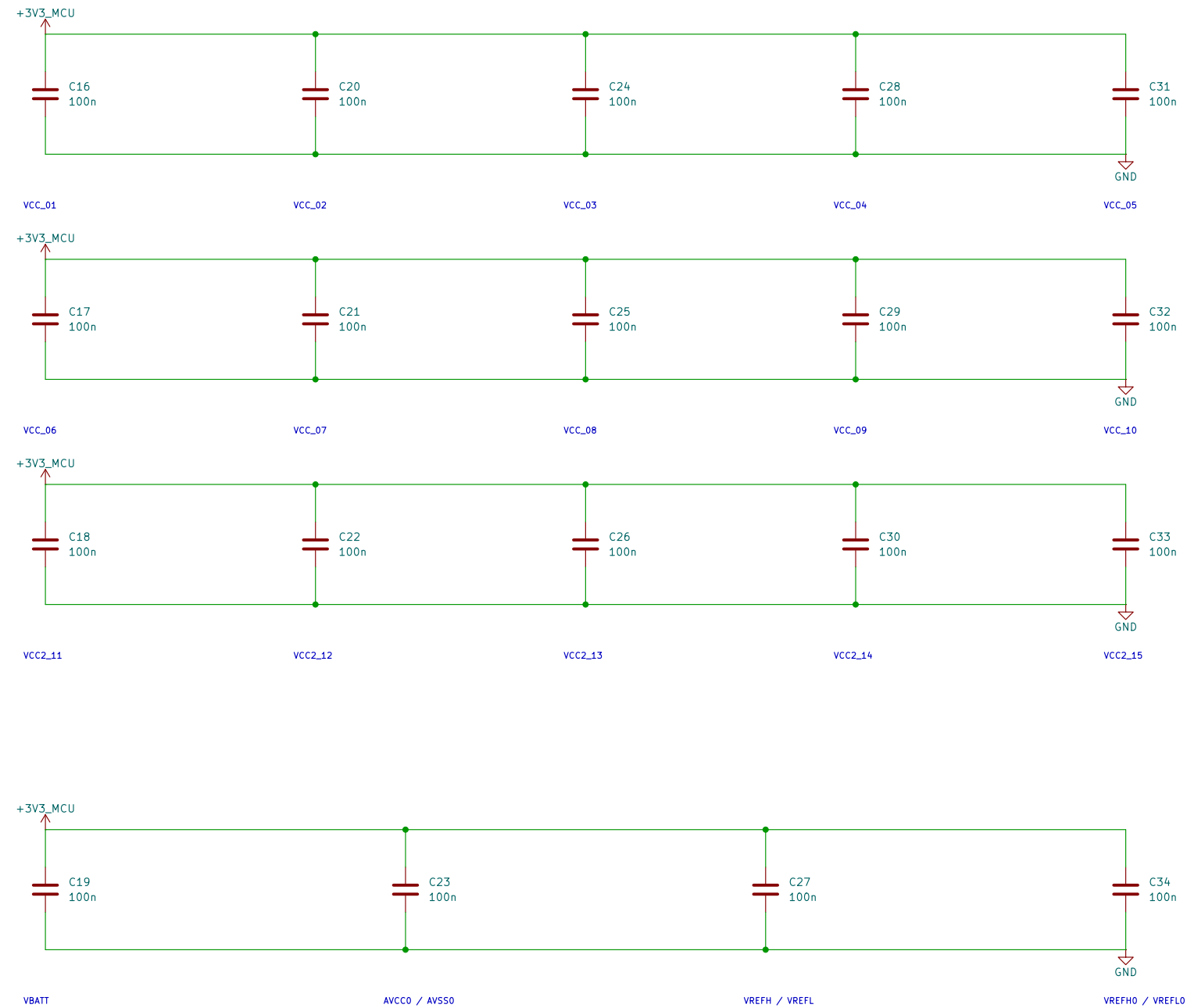
PWR_FLAG

ERC source: internal buck through L1.

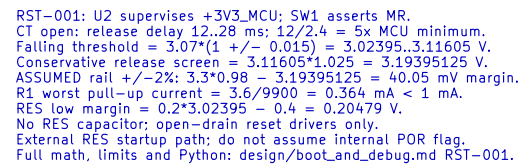
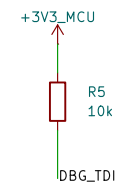
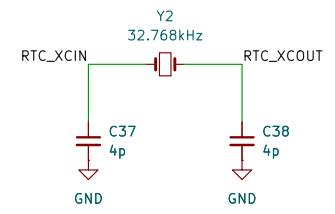
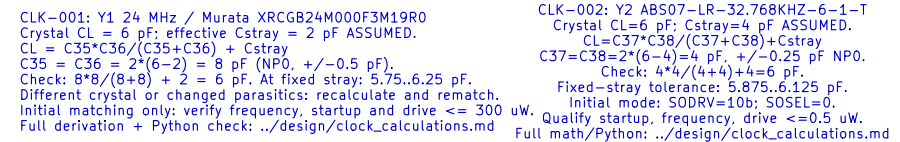




I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



VCC and VCC2 use 3.3 V. All returns use one board ground plane. VBATT follows the main rail; no separate backup-cell supply. Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs. USB/MIP1 supplies remain on the interface sheet and are not yet wired. HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862. This is an electrical draft; capacitor MPN qualification remains open.



References: RA8P1 HUM Rev.1.30 sections 2.5, 4.3–4.4, 21.4.
Quick Design Guide R01AN7883EU0110 sections 2 and 6.
Reset timing: RA8P1 Datasheet Rev.1.30 Table 2.52 p.118.
Exact purchasing selections and status are maintained in the BOM.

A

B

C

D

E

F

A

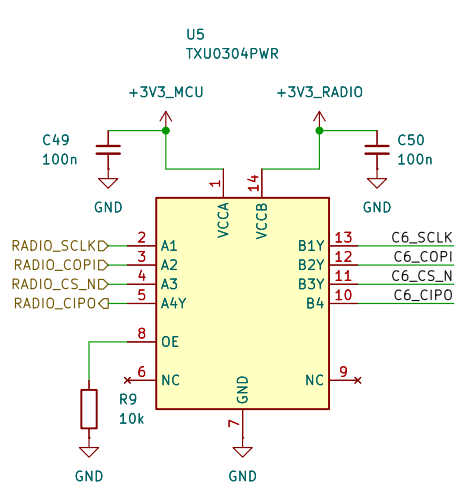
B

C

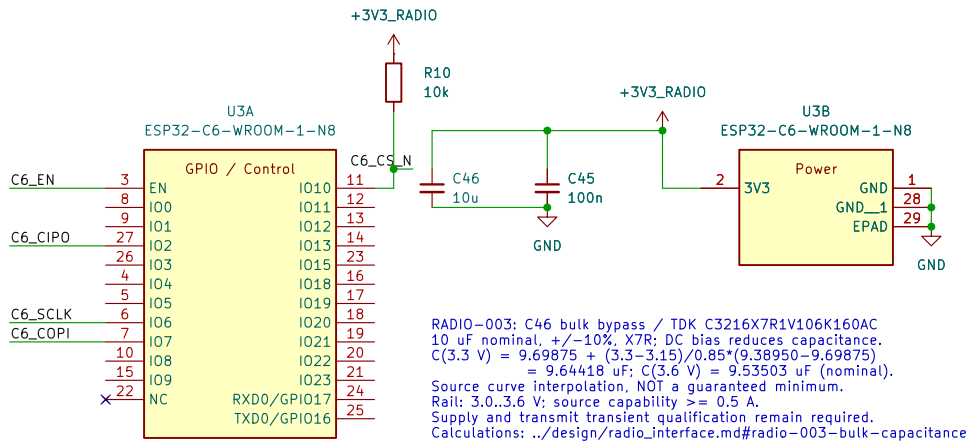
D

E

F

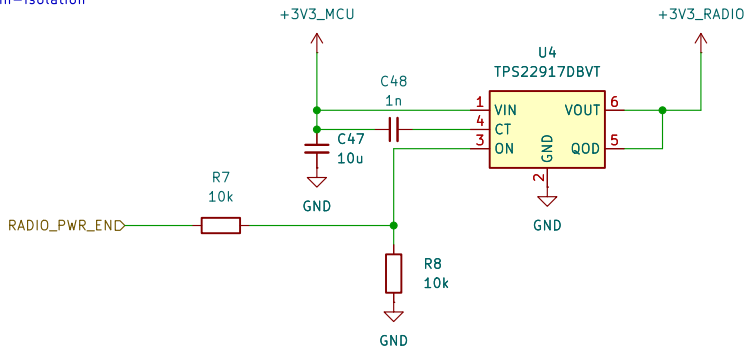


RADIO-009: U5 isolates host and switched-radio signal domains.
C49/C50: 100 nF local bypass; TI TXU0304 section 12.1.
C50 adds 0.1 uF: fitted external radio capacitance = 10.2 uF nominal.
5 MHz budget: 100 - 11 - 11 = 78 ns remaining (5 pF test load).
SPI/OE and reset qualification remain open; timing is not closed.
Math: ../design/radio_interface.md#radio-009-spi-power-domain-isolation



RADIO-003: C46 bulk bypass / TDK C3216X7R1V106K160AC
10 uF nominal, +/-10%, X7R; DC bias reduces capacitance.
 $C(3.3\text{ V}) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)$
 $= 9.64418\text{ uF}$; $C(3.6\text{ V}) = 9.53503\text{ uF}$ (nominal).
Source curve interpolation, NOT a guaranteed minimum.
Rail: 3.0..3.6 V; source capability >= 0.5 A.
Supply and transmit transient qualification remain required.
Calculations: ../design/radio_interface.md#radio-003-bulk-capacitance

RADIO-010: R9/R10 idle-state bias, 10k 1%
 $R_{min}/max = 10k*(1 +/-1%)*(1 +/-1%) = 9801/10201\text{ ohm}$.
12 uA adverse leakage is an ALLOCATION, not measured.
R9: $V_{low} = 12u*10201 = 0.122412\text{ V}$.
TI VT- min = 0.17 V at the 1.1 V equal-rail test point.
R10 @3.3 V: $V_{CS} >= 3.177588\text{ V} > V_{IH}\text{ }2.475\text{ V}$ (25 C).
R9 enabled current = $3.3/10k = 0.330\text{ mA}$.
R10 draws similar current only while CS is low.
OE arbitration, all-state leakage and timing remain open.
Math: ../design/radio_interface.md#radio-010-idle-state-bias



RADIO-004: U4 TPS22917DBVT / radio power gating
QOD tied to VOUT; $R_{dis} = 150\text{ ohm}$ TYPICAL.
 $C45+C46+C50 = 0.1+10+0.1 = 10.2\text{ uF}$ nominal external load.
 $t_{90..10} = \ln(9)*150*10.2u = 3.362\text{ ms}$ (ideal RC).
Not a guaranteed off-time: module C/load and VIN decay matter.
C48 connects CT to VIN, not GND; 50 V rating >= 7 V required.
Reset/isolation and upstream regulator remain incomplete.
Load update: RADIO-009; device limits: RADIO-004.
Math: ../design/radio_interface.md#radio-009-spi-power-domain-isolation

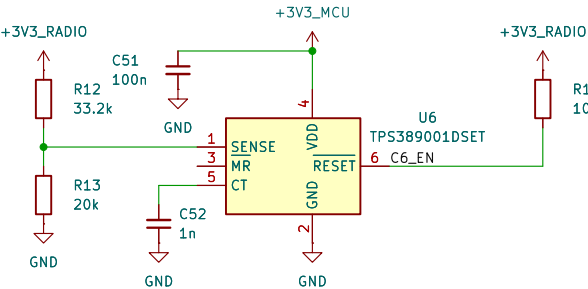
RADIO-005: C47 / U4 VIN local bypass
10 uF nominal, +/-10%, 35 V X7R; same exact TDK part as C46.
 $C(3.3\text{ V}) = 9.69875 + (3.3-3.15)/0.85*(9.38950-9.69875)$
 $= 9.64418\text{ uF}$ nominal; NOT a guaranteed minimum.
Droop = $\text{integral}((I_{load}-I_{source})dt / C_{eff} + \text{ESR/ESL effects})$.
TI section 11: 1 uF usually sufficient; slow sources may need more.
C47 is upstream: exclude it from C45+C46 output-discharge math.
Regulator/transient qualification required; no source flag added.
Calculations: ../design/radio_interface.md#radio-005-input-bypass

RADIO-006: R7/R8 / active-high enable, default OFF
R7 series = R8 pulldown = 10k 1%; +/-2% screening envelope.
TPS22917 ON: low <=0.35 V; high >=1.0 V.
Assume node leakage <=10 uA and host VOL<=0.5 V, VOH>=2.5 V.
 $R_{parallel} <=5.1k$; divider fraction = 0.49..0.51.
 $V_{low} <=0.5*0.51+10u*5.1k = 0.306\text{ V}$ <0.35 V.
 $V_{high} >=2.5*0.49-10u*5.1k = 1.174\text{ V}$ >1.0 V.
HI-Z screen: $10u*10.2k = 0.102\text{ V}$; disable host pull-up.
Host pin, all power states, leakage and timing need qualification.
Full math: ../design/radio_interface.md#radio-006-enable-divder

RADIO-013: R11 / C6_EN reset pull-up
10k 1%, 100ppm/C; $\Delta T=100\text{ C}$ allocation.
 $R_{min}/max=10k*(1 +/-1%)*(1 +/-1%)=9801/10201\text{ ohm}$.
Allocate +/-12uA total adverse node leakage.
 $I_{sink} <=3.6/9801+12uA=0.379310\text{ mA}$ <0.4mA.
U6 VOL<=0.25V at 0.4mA, VDD>=1.5V.
At C6 table point 3.3V, 25 C:
 $V_{EN} >=3.3-12uA*10201=3.177588\text{ V}$ >2.475V VIH.
 $V_{IL}=0.825\text{ V}$; low margin=0.825-0.25=0.575V.
 $P_{max}=3.6^2/9801=1.322314\text{ mW}$.
MR arbitration, power states and timing remain open.
Math: ../design/radio_interface.md (RADIO-013)

RADIO-007: C48 / CT slew-rate control, TDK C1608NP01H102J080AA
1nF +/-5%, 50 V NPO; CT-to-VIN, NOT CT-to-GND.
TI 3.6 V / 25 C TYPICAL table point, not guaranteed at 3.3 V:
Slew = $1900/1000 = 1.9\text{ mV/us}$; $t_{ON} = 3.8*1000 = 3.8\text{ ms}$.
 $t_R(10..90\%) = 1.6*1000 = 1.6\text{ ms}$ (TI test load).
 $C45+C46+C50 = 10.2\text{ uF}$; $I_{cap} = C*dV/dt = 19.38\text{ mA}$.
Open-CT comparison: $10.2u*44\text{ mV/us} = 448.8\text{ mA}$ (typical).
Total inrush also includes module C/load and future rail parts.
Not a current limiter or reset delay; startup qualification required.
Timing model: RADIO-007; updated load: RADIO-009.
Full math: ../design/radio_interface.md#radio-009-spi-power-domain-isolation

RADIO-012: C52 / U6 reset-release delay
C52 = 1nF +/-5% NPO, CT to GND (not VIN).
At 100 C excursion, +/-30 ppm/C: $947.15..1053.15\text{ pF}$.
 $t_{nom} = 1nF*1.23V/1.15uA + 25us = 1.094565\text{ ms}$.
Allocate +/-10nA external CT leakage; start discharged.
 $t_{charge_min} = 947.15pF*1.17V/1.36uA = 814.828\text{ us}$.
 $t_{charge_max} = 1053.15pF*1.29V/0.89uA = 1526.476\text{ us}$.
Charge-only screen, NOT total reset min/max limits.
C6 requires 50us stable supply and 50us reset low.
Complete reset sequencing remains open.
Math: ../design/radio_interface.md#radio-012-supervisor-reset-release-delay



RADIO-014: R12/R13 switched-radio sense divider
33.2k / 20k; Susumu RG, 0.1%, 25ppm/C.
Allocate $\Delta T=100\text{ C}$, 0.15% drift, +/-150nA total.
 $f_{min}/max=(1 +/-0.001)*(1 +/-0.0025)*(1 +/-0.0015)$
 $= 0.99500774825 / 1.00500775375$.
 $V_{trip} = V_{sense}*(1+R_t/R_b) + I_s*R_t$.
 $V_{fall} >=1.15*0.99*(1+R_{tmin}/R_{bmax})-150nA*R_{tmax}$
 $= 3.004600\text{ V}$; margin above 3V only 4.600mV.
 $V_{rise} <=1.15*1.01*1.00825*(1+R_{tmax}/R_{bmin})$
 $+150nA*R_{tmax} = 3.139622\text{ V}$.
PWR-002 release headroom: 14.923mV, static only.
 $I_{div_nom}=3.3/53.2k=62.030uA$; each P <0.652mW.
MR arbitration and brownout dynamics remain open.
Math: ../design/radio_interface.md (RADIO-014)

PWR-002: upstream regulator / reset voltage budget
TPS63802 candidate: +/-1% feedback applies in PWM mode.
Example 56k/10k, 0.1%, 25 ppm/C; $\Delta T=100\text{ C}$.
 $V_{out} = V_{fb}*(1+R_t/R_b) + I_b*R_t$; allocate +/-100 nA.
PWM static screen: $3.24204..3.358485\text{ V}$.
Radio: $3.24204u = 0.5\text{ A}$ *0.175 ohm = 3.154544 V.
Above U6 RADIO-014 rising screen: 14.923 mV remains.
MCU reset release headroom: 48.093 mV.
Not a transient guarantee; regulator circuit not fitted.
Math: ../design/power_decoupling.md (PWR-002)

Incomplete electrical design – not for manufacture

Sheet: /ESP32-C6 radio/
File: radio_esp32.kicad_sch

Title: ESP32-C6 radio

Size: A3 Date: 2026-09-05

Rev: DRAFT

KiCad E.D.A. 10.0.5 Id: 6/6